

# Consumer-Conditional Policy Reversal: Definitions, Criteria, and the Diagnostic

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Two allocation policies can cost the same and still receive opposite class-conditional safety rankings. This document defines that reversal and gives the criteria that turn it into a prospective policy-selection diagnostic. Propositions 2 and 3, with the directional exposure gate of Definition 4, support the following procedure: measure the two directional sensitivities and their floors on calibration data, register the predicted signs and the predicted crossover weight  $\lambda^*$ , then let a graded A/B run confirm or refute the registration. The operator’s deliverable is a selection: which policy to deploy given a class mixture, read off from the position of the fleet’s mixture weight relative to the measured  $\lambda^*$ . The worked instances in Section 5 demonstrate the retrospective half of this procedure on two sealed records, one optical and one radio. The registered-prediction cells that exercise the prospective half are new sealed work; this document supplies the formal basis for their registered predictions.

## 1 Definitions

**Definition 1** (Allocation, cost, resource equivalence). An allocation is a vector  $m \in \mathbb{R}^n$  of per-service margins or resources. A cost functional is a continuously differentiable  $C : \mathbb{R}^n \rightarrow \mathbb{R}$ , strictly increasing in each coordinate. Policies  $m^A, m^B \in \mathbb{R}^n$  are *resource-equivalent* when  $C(m^A) = C(m^B)$ . Write  $d = m^A - m^B$  and  $\bar{m} = \frac{1}{2}(m^A + m^B)$ .

Resource equivalence must be stated in  $C$ , not in nominal units. Equal mean margin in dB is the nominal statement,  $C$  linear in  $m$ . When the true cost (capacity, goodput) is nonlinear in dB, the budget-neutral direction is characterized exactly by the integrated gradient. By the fundamental theorem of calculus,

$$C(m^A) - C(m^B) = \bar{q}^\top d, \quad \bar{q} = \int_0^1 \nabla C(m^B + t d) dt, \quad (1)$$

so  $m^A$  and  $m^B$  are resource-equivalent if and only if  $\bar{q}^\top d = 0$ , with  $\bar{q}$  the integrated gradient. The operating-point gradient  $q = \nabla C(\bar{m})$  satisfies  $q^\top d = 0$  only to first order; by Lemma 1 below, applied to  $C$ , the gap is bounded by  $(L_C/4)\|d\|^2$  when  $\nabla C$  is  $L_C$ -Lipschitz on the segment.

**Definition 2** (Class risk and reversal). For consumer class  $c \in \{1, 2\}$ , the class risk is a function  $R_c : \mathbb{R}^n \rightarrow [0, 1]$ , continuously differentiable on a neighborhood of the segment  $[m^B, m^A]$ . Write

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\*Every number quoted in this document is printed by the companion script `flip.theorem.checks.py` from the graded records of the sealed cells XPROTO-QOT-FLIP and XPROTO-CSI-FLIP (seeds 20260827–20260829). The script also verifies each proposition numerically. All checks pass.

$\Delta R_c = R_c(m^A) - R_c(m^B)$ . The resource-equivalent pair  $(m^A, m^B)$  exhibits *consumer-conditional policy reversal* when

$$\Delta R_1 < 0 < \Delta R_2 \quad \text{or} \quad \Delta R_2 < 0 < \Delta R_1, \quad (2)$$

that is, class 1 is strictly safer under one policy and class 2 strictly safer under the other.

**Definition 3** (Read operator). In the systems studied here each class risk has the form of an exceedance or shortfall probability: the consumer applies a map from system state to a scalar (its *read operator*) and compares that scalar to its threshold. Two classes are *read-aligned* when they apply the same map and differ at most in the threshold; otherwise their reads are *misaligned*. The definitions above do not require this structure; it is named because Section 3 and the diagnostic refer to it.

## 2 Reversal criteria

**Proposition 1** (Exact directional form). *For each class  $c$  there exists  $\xi_c$  on the open segment  $(m^B, m^A)$  with*

$$\Delta R_c = \nabla R_c(\xi_c)^\top d. \quad (3)$$

*Hence reversal holds if and only if the two directional derivatives, each evaluated at its own mean value point, have opposite signs:  $(\nabla R_1(\xi_1)^\top d)(\nabla R_2(\xi_2)^\top d) < 0$ .*

*Proof.* Apply the mean value theorem to  $f_c(t) = R_c(m^B + t d)$  on  $[0, 1]$ ;  $f_c$  is differentiable there because  $R_c$  is  $C^1$  on a neighborhood of the segment, and  $f'_c(t) = \nabla R_c(m^B + t d)^\top d$ .  $\square$

This criterion is exact and unmeasurable. The points  $\xi_1, \xi_2$  are unknown, differ between the classes, and are not observable from data. The measurable object is the directional derivative at the midpoint, and it carries an error bound.

**Lemma 1** (Midpoint bound; the constant is  $1/4$ ). *Suppose  $\nabla R_c$  is  $L_c$ -Lipschitz on the segment  $[m^B, m^A]$ . Then*

$$|\Delta R_c - \nabla R_c(\bar{m})^\top d| \leq \frac{L_c}{4} \|d\|^2. \quad (4)$$

*The constant  $1/4$  is sharp in this class. If in addition  $R_c$  is twice differentiable along the segment,  $L_c \|d\|^2$  may be replaced by the directional curvature bound  $M_c = \sup_{t \in [0, 1]} |d^\top \nabla^2 R_c(m^B + t d) d| \leq L_c \|d\|^2$ , giving the sharper bound  $M_c/4$ .*

*Proof.* With  $f_c(t) = R_c(m^B + t d)$ ,

$$\Delta R_c - \nabla R_c(\bar{m})^\top d = \int_0^1 (f'_c(t) - f'_c(\tfrac{1}{2})) dt, \quad |f'_c(t) - f'_c(\tfrac{1}{2})| \leq L_c \|d\|^2 |t - \tfrac{1}{2}|,$$

and  $\int_0^1 |t - \tfrac{1}{2}| dt = \tfrac{1}{4}$ . The directional version is the same computation with  $|f'_c(t) - f'_c(\tfrac{1}{2})| \leq M_c |t - \tfrac{1}{2}|$ . Sharpness: take  $n = 1$ ,  $R(x) = \frac{L}{2}(x - c)|x - c|$ , whose derivative  $R'(x) = L|x - c|$  is  $L$ -Lipschitz, and centre the segment at  $c$ . Then  $f'_c(t) = L\|d\|^2 |t - \tfrac{1}{2}|$ , the midpoint term vanishes, and the error equals  $(L/4)\|d\|^2$  exactly. The companion script verifies the bound on random quadratic, Gaussian-shortfall, and sinusoidal families and reproduces the equality instance to machine precision.  $\square$

*Remark 1* (Why not 1/8). A per-side Taylor expansion from the midpoint bounds each half-step error by  $\frac{L_c}{2} \|d/2\|^2 = (L_c/8) \|d\|^2$ , which suggests 1/8. The two half-step errors add in the worst case, and the equality instance above attains  $(L_c/4) \|d\|^2$ , strictly violating a 1/8 bound. The correct constant is 1/4. For quadratic  $R_c$  the error is exactly zero:  $f'_c$  is then linear and the midpoint value integrates it exactly. Near-tightness therefore cannot be exhibited on a quadratic; the script exhibits exactness on quadratics and equality on the piecewise-quadratic instance.

**Proposition 2** (Predictive criterion). *Let  $g_c = \nabla R_c(\bar{m})^\top d$  and suppose  $\nabla R_c$  is  $L_c$ -Lipschitz on the segment. If*

$$|g_c| > \frac{L_c}{4} \|d\|^2 \quad \text{for } c = 1, 2, \quad (5)$$

*then  $\text{sign}(\Delta R_c) = \text{sign}(g_c)$  for both classes. Consequently, if in addition  $g_1 g_2 < 0$ , reversal holds.*

*Proof.* By Lemma 1,  $\Delta R_c$  lies within  $(L_c/4) \|d\|^2$  of  $g_c$ ; under (5) this interval excludes zero, so the signs agree. Opposite signs of  $g_1, g_2$  then give (2).  $\square$

The condition (5) is directional, and it has to be. The quantity that predicts the reversal is the derivative of each class risk along the specific budget-neutral direction  $d$ , not the size of the gradient: a large gradient orthogonal to  $d$  produces no policy difference at all. We therefore define exposure as the directional object.

**Definition 4** (Exposure and the exposure gate). The *exposure* of class  $c$  to the policy pair is

$$E_c(d) = |\nabla R_c(\bar{m})^\top d|. \quad (6)$$

Operationally,  $\nabla R_c(\bar{m})^\top d$  is estimated by symmetric finite differences along  $d$  on calibration seeds; one scalar per class, with standard error  $\text{SE}_c$  across seeds. The registered prediction is licensed when

$$E_c(d) > z \text{SE}_c + \frac{L_c}{4} \|d\|^2 \quad \text{for both classes,} \quad (7)$$

where  $z$  is the declared confidence multiplier and the second term is the Taylor-remainder bound of Lemma 1 (or its directional refinement  $M_c/4$  when a curvature bound along  $d$  is available). Equivalently, both dimensionless exposure ratios  $E_c(d) / (z \text{SE}_c + (L_c/4) \|d\|^2)$  must exceed one.

The gate is where near-threshold operation enters. Near-threshold operation is what makes the directional derivative along the policy difference large enough to clear the noise-plus-curvature floor. Far from threshold,  $E_c(d)$  sinks below the floor even if  $\|\nabla R_c\|$  is large in some other direction, and the gate withholds the prediction (Section 3(b) exhibits both failure modes).

**Proposition 3** (Mixture crossover; exact). *For fleet weight  $\lambda \in [0, 1]$  on class 1, let  $R_\lambda = \lambda R_1 + (1 - \lambda) R_2$ . Then, with no linearization,*

$$\Delta R_\lambda = \lambda \Delta R_1 + (1 - \lambda) \Delta R_2. \quad (8)$$

*$\Delta R_\lambda$  changes sign on  $(0, 1)$  if and only if reversal (2) holds, and then the crossover is unique:*

$$\lambda^* = \frac{\Delta R_2}{\Delta R_2 - \Delta R_1} \in (0, 1), \quad \Delta R_\lambda = (\lambda - \lambda^*) (\Delta R_1 - \Delta R_2). \quad (9)$$

*Proof.* Equation (8) is linearity of the difference in  $\lambda$ . The map  $\lambda \mapsto \Delta R_\lambda$  is affine with endpoint values  $\Delta R_2$  at  $\lambda = 0$  and  $\Delta R_1$  at  $\lambda = 1$ ; it has a zero in the open interval iff the endpoint values have strictly opposite signs, which is (2), and an affine function has at most one zero. Solving  $\lambda \Delta R_1 + (1 - \lambda) \Delta R_2 = 0$  gives (9). The script confirms the identity and the equivalence on  $10^5$  random draws.  $\square$

**Corollary 1** (No mixture-free scalar ranking). *Under reversal, the two classes order the policy pair oppositely, so no scalar summary of the pair reproduces both class verdicts. Within the mixture family,  $R_\lambda$  ranks  $A$  safer than  $B$  exactly when  $\lambda$  is on one side of  $\lambda^*$  (the side with  $(\lambda - \lambda^*)(\Delta R_1 - \Delta R_2) < 0$ ). Every fleet aggregate therefore orders the pair by an implicit choice of  $\lambda$  relative to  $\lambda^*$ ; the aggregate’s verdict is a property of the mixture, not of the policies alone.*

### 3 Scope: sufficiency, not necessity

The mechanism studied in this paper is misaligned reads combined with exposure above the gate floor. Proposition 4 bounds what that mechanism explains: it is sufficient for the systems studied and predictive there, and it is not a characterization of reversal in general. Both directions fail, constructively.

**Proposition 4** (Neither direction is an implication). (a) Reversal without misalignment. *There is a resource-equivalent pair, a single read operator shared by both classes, and thresholds  $\tau_1 \neq \tau_2$  such that reversal holds.* (b) Misalignment without reversal. *There are misaligned read operators and resource-equivalent pairs with no reversal: (b1) with both gradients large in norm but orthogonal to  $d$ , so that  $\Delta R_1 = \Delta R_2 = 0$  exactly; (b2) with both classes far from threshold, so that both deltas are negligible and share a sign.*

*Construction, verified numerically in the companion script.* (a) Take  $n = 2$ ,  $C(m) = m_1 + m_2$ ,  $m^A = (1, 1)$ ,  $m^B = (0, 2)$ , so  $C(m^A) = C(m^B) = 2$  and  $d = (1, -1)$ . Let the outcome read by both classes be  $Y \sim \mathcal{N}(\mu(m), \sigma(m)^2)$  with  $\mu(m) = (m_2 - 1)/2$  and  $\sigma(m) = 2 - m_1$ : under  $A$ ,  $Y \sim \mathcal{N}(0, 1)$ ; under  $B$ ,  $Y \sim \mathcal{N}(0.5, 2^2)$ . Both classes apply the identical read operator (read  $Y$ ) and differ only in threshold; class  $c$  fails when  $Y < \tau_c$ , with  $\tau_1 = -2$  and  $\tau_2 = 1$ . The two shortfall curves cross:  $R_1(A) = \Phi(-2) = 0.0228$ ,  $R_1(B) = \Phi(-1.25) = 0.1056$ ,  $R_2(A) = \Phi(1) = 0.8413$ ,  $R_2(B) = \Phi(0.25) = 0.5987$ , so  $\Delta R_1 = -0.0829 < 0 < \Delta R_2 = +0.2426$ : reversal, with  $\lambda^* = 0.7453$ . The exposure gate licenses this instance: the midpoint directional derivatives are  $-0.0863$  and  $+0.2347$ , and the exposure ratios against the directional curvature floors are 4.27 and 3.37.

(b1) Take  $n = 3$ ,  $C(m) = m_1 + m_2 + m_3$ ,  $d = (0.5, -0.5, 0)$ ,  $m^B = 0$ , and shortfall risks  $R_c(m) = \Phi(\tau - w_c^\top m)$  with  $w_1 = (1, 1, 0.5)$ ,  $w_2 = (1, 1, -0.5)$ ,  $\tau = 0$ . The reads are misaligned and the operating point sits at threshold: both risks equal 0.5000 and both gradient norms equal 0.5984. But  $w_1^\top d = w_2^\top d = 0$ , so  $E_1(d) = E_2(d) = 0$  and  $\Delta R_1 = \Delta R_2 = 0$  exactly. Large gradients, no policy difference: the directional exposure  $E_c(d)$ , not the gradient norm, is the diagnostic quantity.

(b2) Take  $n = 2$ , the same cost,  $w_1 = (1, 0.3)$ ,  $w_2 = (1, 0.5)$ ,  $m^B = (3, 3)$ ,  $d = (0.5, -0.5)$ : about four standard deviations from threshold. The four risks are  $1.07 \times 10^{-5}$ ,  $4.81 \times 10^{-5}$ ,  $1.02 \times 10^{-6}$ ,  $3.40 \times 10^{-6}$  ( $R_1(A), R_1(B), R_2(A), R_2(B)$ ); both deltas are negative, so there is no reversal, and the exposures  $3.46 \times 10^{-5}$  and  $2.26 \times 10^{-6}$  sit below their gate floors 0.0330 and 0.0378 by factors of about  $10^3$  and  $10^4$ . No prediction would be registered.  $\square$

The conclusion is stated exactly at its strength. Misaligned reads plus near-threshold operation is a sufficient and empirically predictive mechanism in the studied systems; it is not a necessary condition for reversal, and Proposition 4(a) shows reversal without misalignment. Statements of the form “reversal occurs when reads are misaligned” are not claimed and are false as implications.

## 4 The reversal diagnostic

The criteria above assemble into a prospective procedure. Its output before the graded run is either a registered, falsifiable prediction or a registered abstention; the gate abstains, it does not hedge. A licensed prediction is falsifiable in the graded run, and an unlicensed configuration yields no registered prediction rather than a weak one. That property keeps the mechanism claim at sufficiency.

### The reversal diagnostic.

1. Declare the two consumer classes, their read operators and thresholds, and the cost functional  $C$ .
2. Verify resource equivalence in  $C$ , not in nominal units:  $C(m^A) = C(m^B)$ , equivalently  $\bar{q}^\top d = 0$  for the integrated gradient  $\bar{q}$  of (1).
3. Measure the scalar directional derivative  $g_c = \nabla R_c(\bar{m})^\top d$  by symmetric finite differences along  $d$  with a declared perturbation size  $\varepsilon$ :  $\hat{g}_c = [\hat{R}_c(\bar{m} + \varepsilon d) - \hat{R}_c(\bar{m} - \varepsilon d)] / (2\varepsilon)$ , replicated on calibration seeds. One scalar per class; this is cheaper and better conditioned than estimating the full gradient, and it is the only quantity the criterion uses. Record the standard error  $SE_c$  across seeds.
4. Estimate or bound the curvature term:  $L_c$  on the segment, or the directional bound  $M_c$ . Compute both floor terms and the exposure ratio  $E_c(d) / (z SE_c + (L_c/4) \|d\|^2)$  for each class, with declared  $z$ .
5. If and only if both exposure ratios exceed one, register the predicted signs of  $(\Delta R_1, \Delta R_2)$  and the predicted crossover  $\lambda^* = \hat{g}_2 / (\hat{g}_2 - \hat{g}_1)$  before the graded run. Otherwise abstain: no registered prediction.
6. Run A/B on disjoint graded seeds. Grade the sign predictions and report the measured  $\lambda^*$  per seed.

Steps 5 and 6 are the sealed-preregistration discipline used throughout this program: predictions are registered against named seeds before the graded run, and the graded record either passes or fails them with no post hoc adjustment.

## 5 Worked instances

Two sealed cells instantiate the definitions, one optical and one radio. All numbers in this section are taken verbatim from the graded records (seeds 20260827–20260829, disjoint from the shakedown seeds) and are printed by the companion script, which also recomputes every

derived quantity. These records predate the diagnostic’s step-3–5 gradient registration; they exercise the retrospective half of the procedure (measured  $\Delta R_c$ , reversal indicator, and  $\lambda^*$ ), and the gradient-registered cells are new sealed work.

**Optical (XPROTO-QOT-FLIP).** The substrate is a GN-model quality-of-transmission simulation (GNPy) over the CORONET CONUS topology with 120 services per seed. Class R holds the long-reach services at band-edge spectral positions, whose penalty is dominated by span count; class P holds the short-reach services at band-centre positions, whose penalty is dominated by nonlinear-interference position. Policy A allocates margin proportional to reach and policy B proportional to band-centrality, both normalized to the same fleet-mean margin of 1.0 dB, and the class risk is the false-clear rate: the fraction of services whose selected format passes on the estimate plus margin but fails the deployed GSNR. Table 1 reports the record. Reversal holds on every seed,  $\Delta R_R < 0 < \Delta R_P$ , with  $\lambda^* \in 0.382\text{--}0.422$  (weight on class R). The recorded fleet false-clear rates at the even mixture are (0.4250, 0.4916), (0.4167, 0.4667), and (0.3750, 0.4333) for  $(A, B)$  across the three seeds; each equals the even mixture of its class rates to the record’s rounding, and each fleet difference is negative, as Proposition 3 requires for  $\lambda = 1/2 > \lambda^*$  with  $\Delta R_1 < \Delta R_2$ .

Table 1: XPROTO-QOT-FLIP, graded record. Risks are false-clear rates;  $\Delta R = R(m^A) - R(m^B)$ ;  $\lambda^*$  is the class-R weight at which the mixture verdict crosses; the last column is the recorded fleet difference at the even mixture.

| seed     | $R_R(A)$ | $R_R(B)$ | $R_P(A)$ | $R_P(B)$ | $\Delta R_R$ | $\Delta R_P$ | $\lambda^*$ | $\Delta R_{1/2}$ |
|----------|----------|----------|----------|----------|--------------|--------------|-------------|------------------|
| 20260827 | 0.3833   | 0.7333   | 0.4667   | 0.2500   | −0.3500      | +0.2167      | 0.382       | −0.0666          |
| 20260828 | 0.3667   | 0.7000   | 0.4667   | 0.2333   | −0.3333      | +0.2334      | 0.412       | −0.0500          |
| 20260829 | 0.3000   | 0.7333   | 0.4500   | 0.1333   | −0.4333      | +0.3167      | 0.422       | −0.0583          |

**Radio (XPROTO-CSI-FLIP).** The substrate is a 5G NR link-adaptation simulation with real Sionna LDPC block-error curves over TDL-A fading, held CSI reports every 20 TTIs, 6000 TTIs per run, and 12 users per fleet. Fleet M (200 Hz Doppler, 12 dB mean SNR) fails through CSI aging, the time axis; fleet S (10 Hz, 7 dB) fails through deep fades, the amplitude axis. Policy A allocates the 1.5 dB fleet-mean margin by Doppler and policy B by SNR deficit, and the class risk is the mean empirical HARQ NACK rate. Table 2 reports the record. Reversal holds on every seed,  $\Delta R_M < 0 < \Delta R_S$ , with  $\lambda^* \in 0.426\text{--}0.432$  (weight on class M). The recorded fleet NACK rates at the even mixture are (0.2452, 0.2631), (0.2409, 0.2571), and (0.2447, 0.2618) for  $(A, B)$ ; the same consistency checks hold, and the even-mixture verdict again sits on the  $\lambda = 1/2 > \lambda^*$  side.

Table 2: XPROTO-CSI-FLIP, graded record. Risks are mean empirical HARQ NACK rates; columns as in Table 1, with  $\lambda^*$  the class-M weight.

| seed     | $R_M(A)$ | $R_M(B)$ | $R_S(A)$ | $R_S(B)$ | $\Delta R_M$ | $\Delta R_S$ | $\lambda^*$ | $\Delta R_{1/2}$ |
|----------|----------|----------|----------|----------|--------------|--------------|-------------|------------------|
| 20260827 | 0.2222   | 0.3601   | 0.2682   | 0.1660   | −0.1379      | +0.1022      | 0.426       | −0.0179          |
| 20260828 | 0.2205   | 0.3564   | 0.2613   | 0.1579   | −0.1359      | +0.1034      | 0.432       | −0.0162          |
| 20260829 | 0.2226   | 0.3597   | 0.2668   | 0.1638   | −0.1371      | +0.1030      | 0.429       | −0.0171          |

**Aligned controls.** Each cell registered a read-aligned control run under the same protocol. In the optical cell the control replaces the two footprint classes with two FEC classes (soft- and hard-decision thresholds, the required table shifted by  $\mp 1.5$  dB) that share the read projection. The control deltas  $(\Delta R_{SD}, \Delta R_{HD})$  are  $(+0.1333, +0.1334)$ ,  $(+0.0500, +0.0334)$ , and  $(+0.1167, +0.0500)$ : same sign on every seed, policy B safer for both classes, no reversal, and the recorded `fec_flip` field is false on every seed. In the radio cell the control reads one fleet-M trace against required-SNR tables shifted by 0 and +2 dB under protect-low versus protect-high margins. The control deltas  $(\Delta R_{t1}, \Delta R_{t2})$  are  $(-0.1126, -0.0736)$ ,  $(-0.1024, -0.0848)$ , and  $(-0.1022, -0.0802)$ : same sign on every seed, policy A safer at both thresholds, no reversal, and `null_flip` is false on every seed. These controls sit on the no-reversal side of Proposition 4: aligned reads whose outcome distributions do not cross on the shared scalar, so both thresholds move together along  $d$ . Proposition 4(a) shows that alignment alone does not forbid reversal; the same-sign outcome here is a measured property of these systems’ distributions, registered in advance as the expected control behavior, and it held on all six control runs.

## 6 Relation to existing formalisms

Reversal under aggregation is not new as a phenomenon. Simpson’s paradox reverses subgroup orderings in the aggregate, and sufficient conditions for its occurrence are established in specific settings [1, 2]. Rank reversal is a known failure mode of multi-criteria decision methods [3]. Under reversal the two policies are Pareto-incomparable, and Proposition 3 locates the scalarization weight at which any mixture ordering changes side. Opposite-sign subgroup effects are the object of the heterogeneous treatment-effect literature [4]. The contribution of this paper is not the existence of reversal in aggregation; it is the preregistered predictive diagnostic on networking substrates: measured directional sensitivities, a declared exposure gate, registered signs and  $\lambda^*$ , and graded outcomes on sealed seeds. Within the program’s own series, this claim is distinct from Paper IV’s compression flip [5]: the compression flip concerns a single consumer against the reconstruction objective over a fixed bit budget, a claim about codes, while the policy reversal here needs two consumer classes over a resource-equivalent allocation pair and yields what a single consumer cannot, opposite class-conditional signs and hence no mixture-free scalar ranking.

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